

Data_Mining_Project

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Loading Libraries

```
library(MASS)
library(ISLR2)
```

```
## Warning: package 'ISLR2' was built under R version 4.4.3
```

```
##
```

```
## Attaching package: 'ISLR2'
```

```
## The following object is masked from 'package:MASS':
```

```
##
```

```
## Boston
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.4.3
```

Loading Boston Data set

```
head(Boston)
```

```
##      crim zn  indus chas   nox    rm  age    dis rad tax ptratio lstat medv
## 1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900   1 296    15.3  4.98 24.0
## 2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671   2 242    17.8  9.14 21.6
## 3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671   2 242    17.8  4.03 34.7
## 4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622   3 222    18.7  2.94 33.4
## 5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622   3 222    18.7  5.33 36.2
## 6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622   3 222    18.7  5.21 28.7
```

```
?Boston
```

```
## starting httpd help server ... done
```

To build model and attach data

```
lm.fit <-lm(medv~lstat, data = Boston)
attach(Boston)
lm.fit <-lm(medv~lstat)
```

to get model fit

```
lm.fit
```

```
##
## Call:
## lm(formula = medv ~ lstat)
##
## Coefficients:
## (Intercept)      lstat
##      34.55      -0.95
```

```
summary(lm.fit)
```

```
##
## Call:
## lm(formula = medv ~ lstat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.168  -3.990  -1.318   2.034  24.500
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  34.55384    0.56263   61.41  <2e-16 ***
## lstat       -0.95005    0.03873  -24.53  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.216 on 504 degrees of freedom
## Multiple R-squared:  0.5441, Adjusted R-squared:  0.5432
## F-statistic: 601.6 on 1 and 504 DF, p-value: < 2.2e-16
```

```
# names of other information stored in model
names(lm.fit)
```

```
## [1] "coefficients" "residuals"      "effects"      "rank"
## [5] "fitted.values" "assign"        "qr"           "df.residual"
## [9] "xlevels"       "call"         "terms"        "model"
```

```
# To get coefficient
```

```
coef(lm.fit)
```

```
## (Intercept)      lstat
## 34.5538409 -0.9500494
```

```
# To get CI
```

```
confint(lm.fit)
```

```
##                2.5 %      97.5 %
## (Intercept) 33.448457 35.6592247
## lstat      -1.026148 -0.8739505
```

```
# To do prediction
```

```
predict(lm.fit, data.frame(lstat = (c(5, 10, 15))),
interval = "confidence")
```

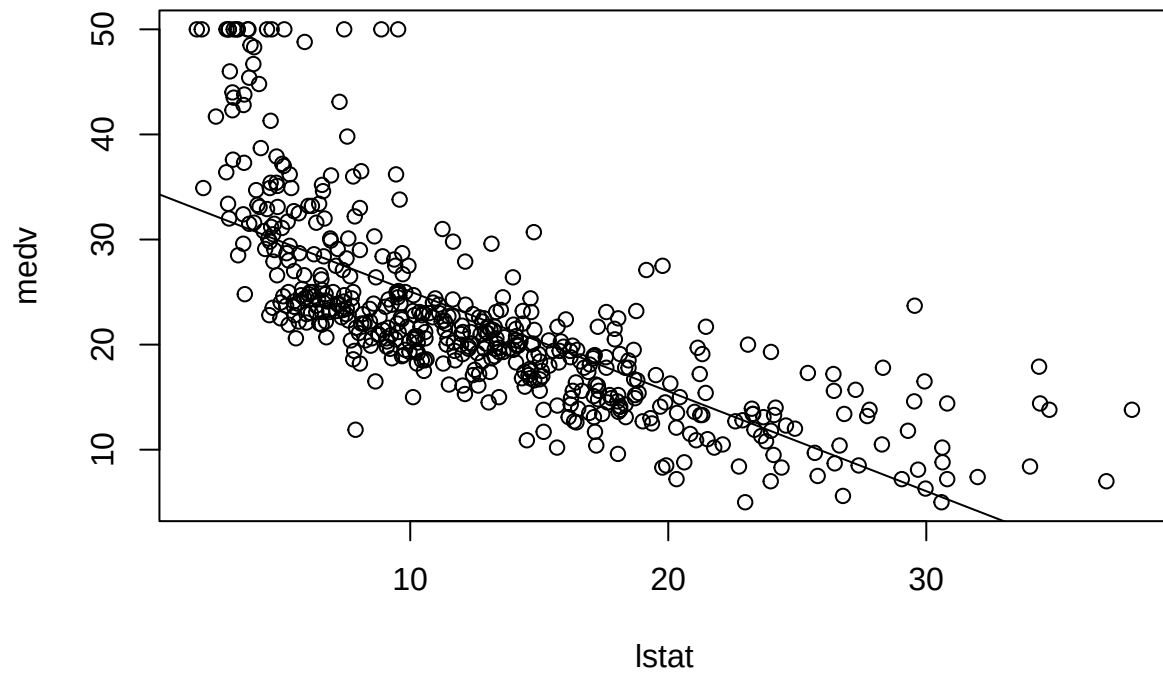
```
##          fit          lwr          upr
## 1 29.80359 29.00741 30.59978
## 2 25.05335 24.47413 25.63256
## 3 20.30310 19.73159 20.87461
```

```
#
predict(lm.fit, data.frame(lstat = (c(5, 10, 15))),
interval = "prediction")
```

```
##          fit          lwr          upr
## 1 29.80359 17.565675 42.04151
## 2 25.05335 12.827626 37.27907
## 3 20.30310  8.077742 32.52846
```

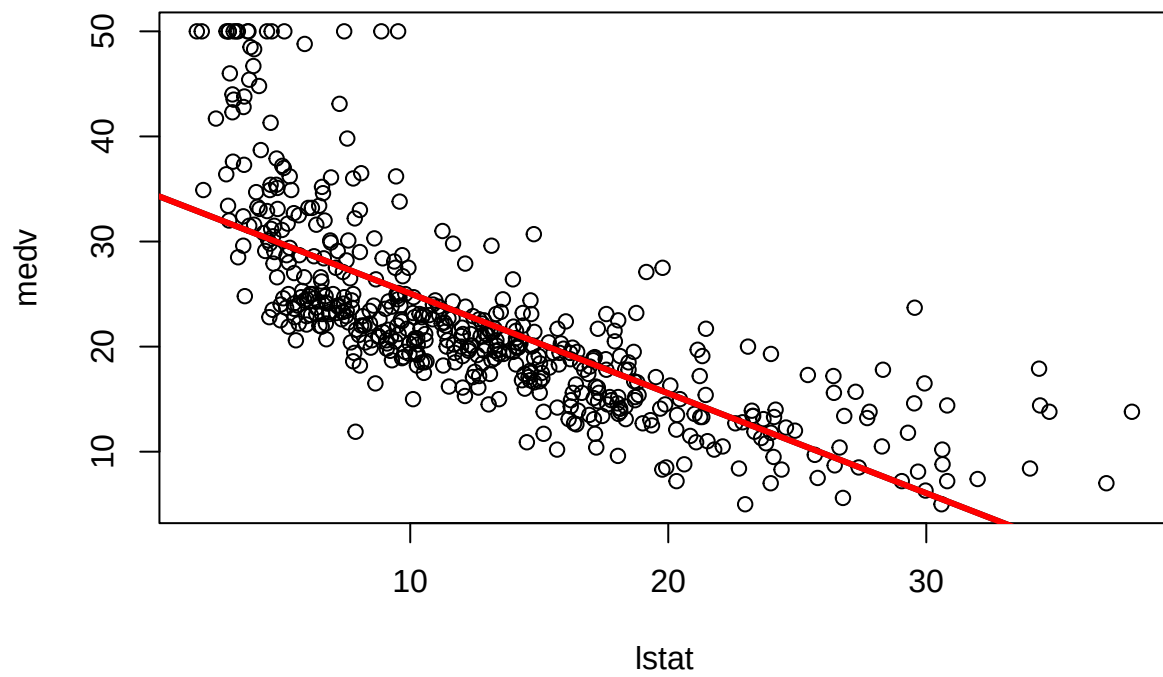
```
# We will now plot medv and lstat along with the least squares regression
#line using the plot() and abline() functions
```

```
plot(lstat, medv)
abline(lm.fit)
```

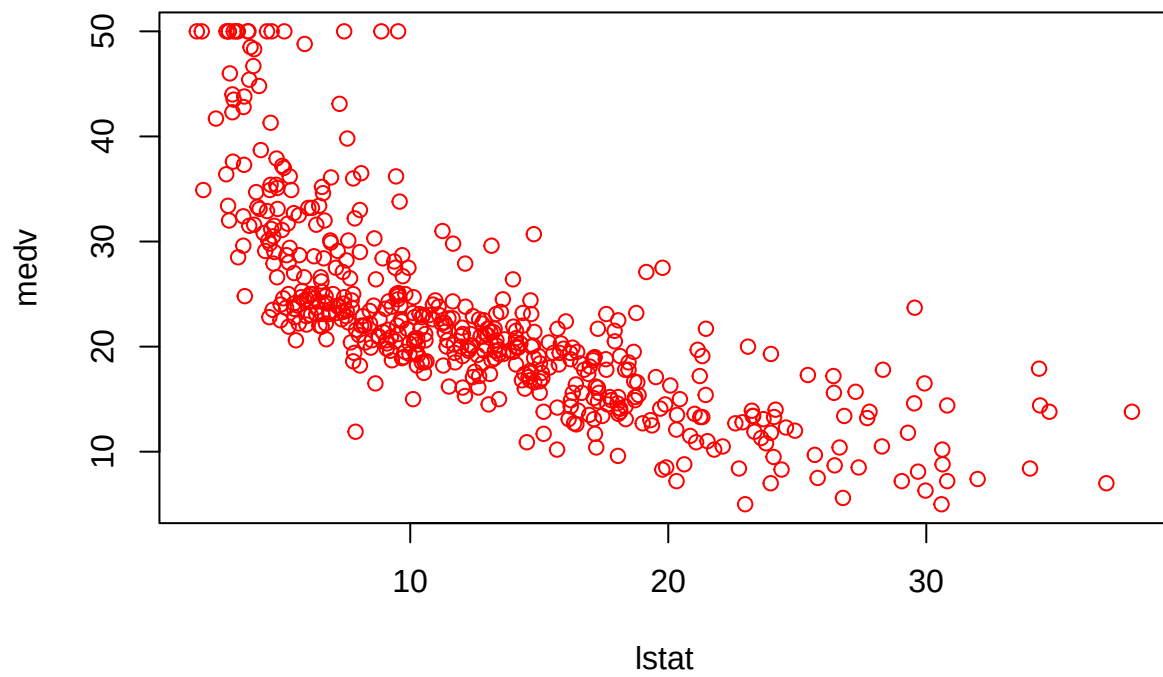


There is some evidence for non-linearity in the relationship between lstat # and medv.

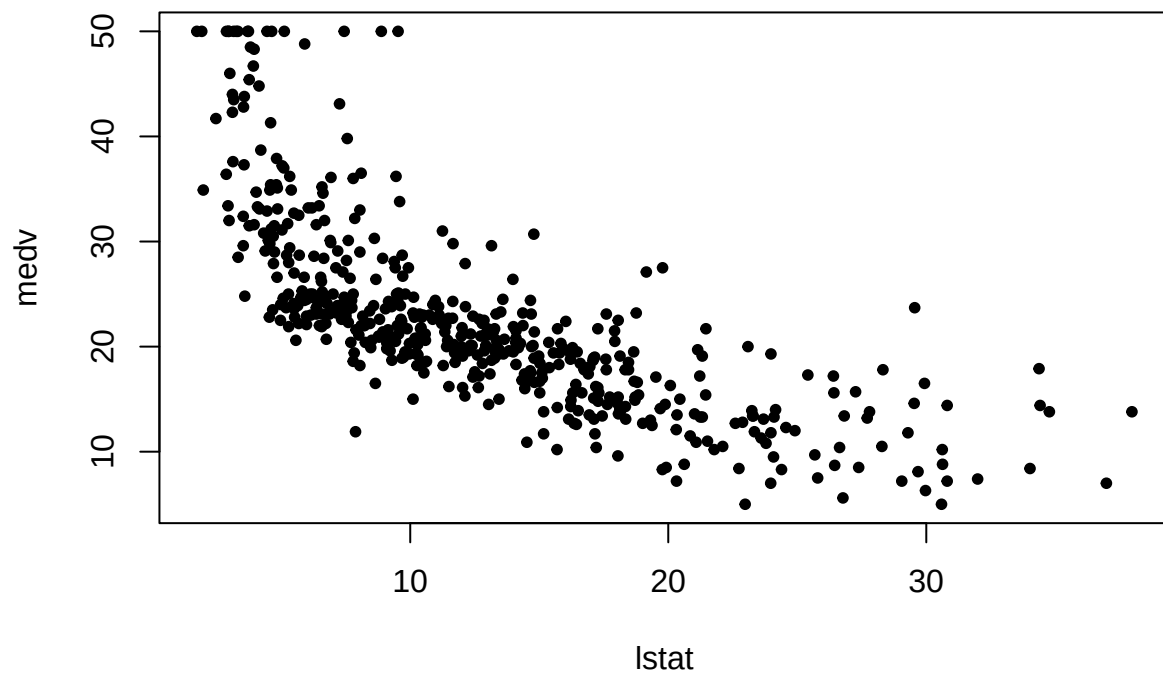
```
plot(lstat, medv)
abline(lm.fit)
abline(lm.fit, lwd = 3)
abline(lm.fit, lwd = 3, col = "red")
```



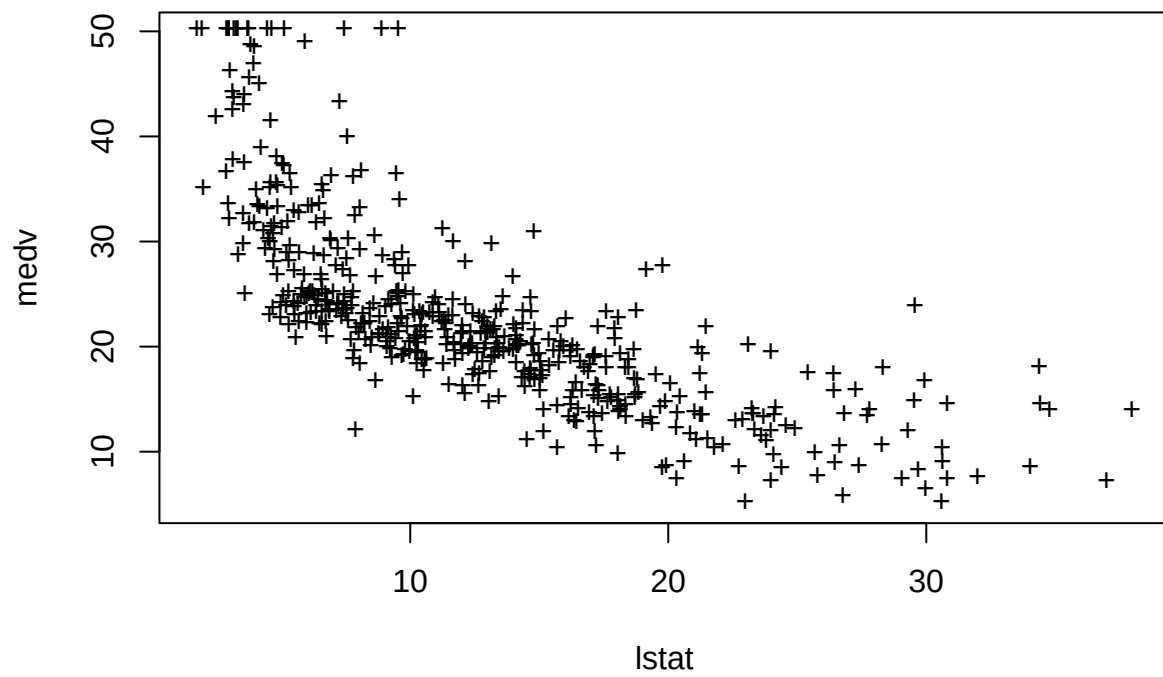
```
plot(lstat, medv, col = "red")
```



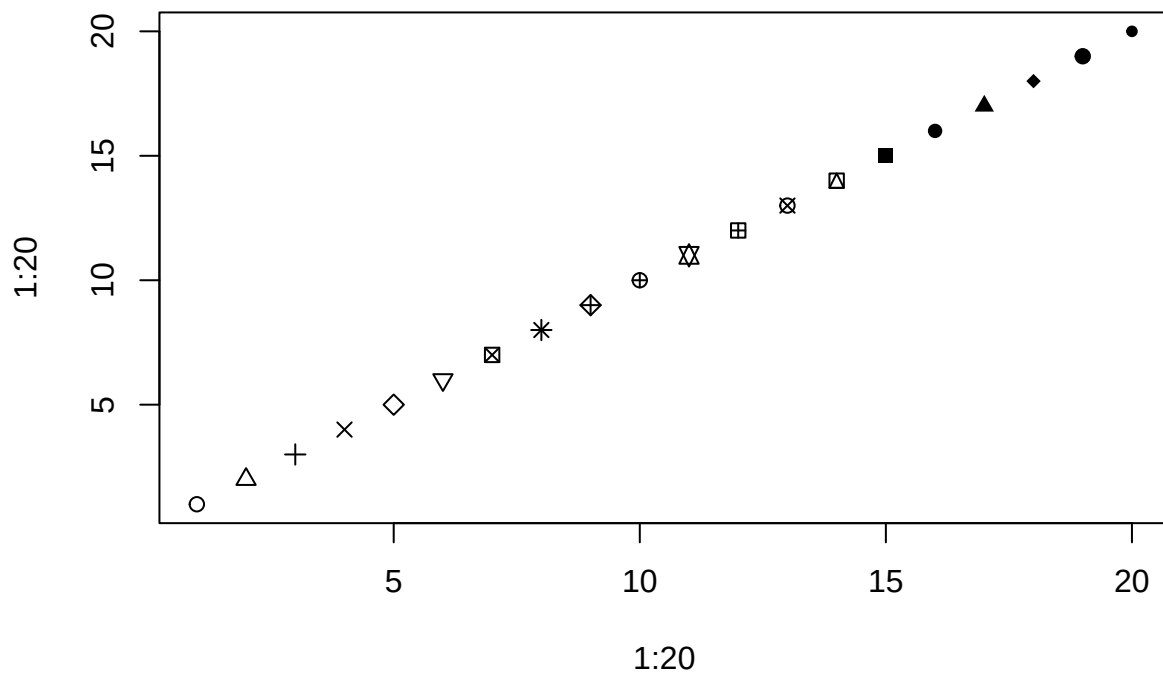
```
plot(lstat, medv, pch = 20)
```



```
plot(lstat, medv, pch = "+")
```

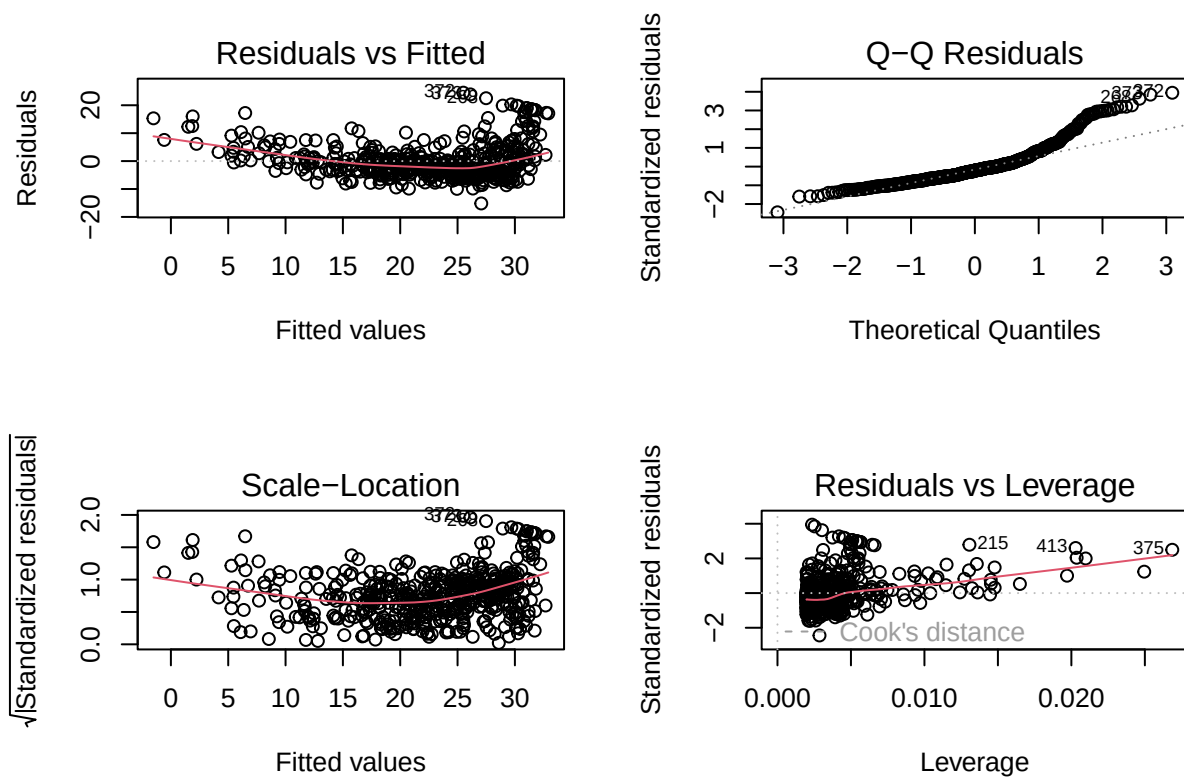


```
plot(1:20, 1:20, pch = 1:20)
```

Diagnostic plots

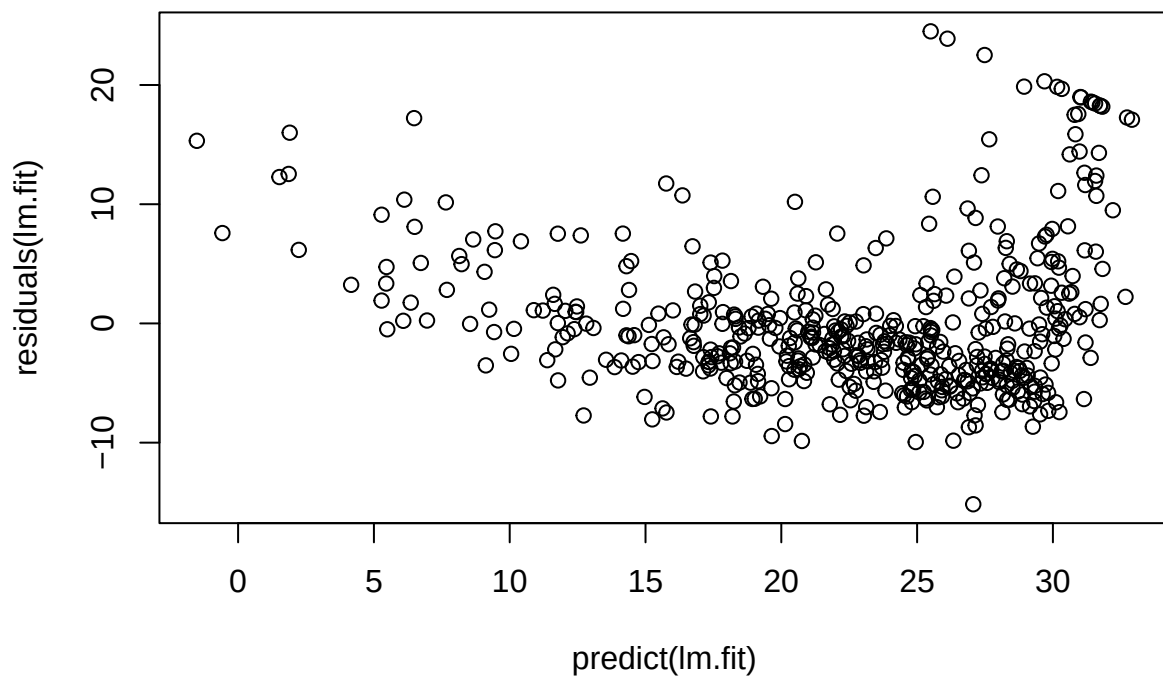
```
par(mfrow = c(2, 2))  
plot(lm.fit)
```



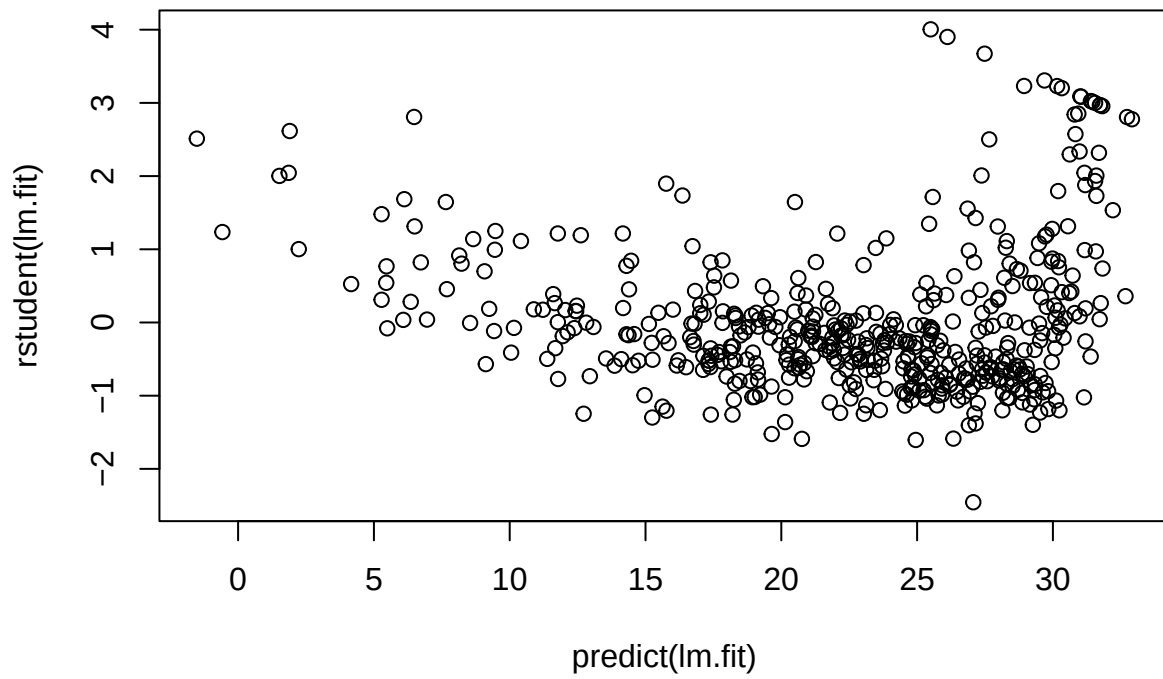
we can compute the residuals from a linear regression fit

using the `residuals()`

```
plot(predict(lm.fit), residuals(lm.fit))
```

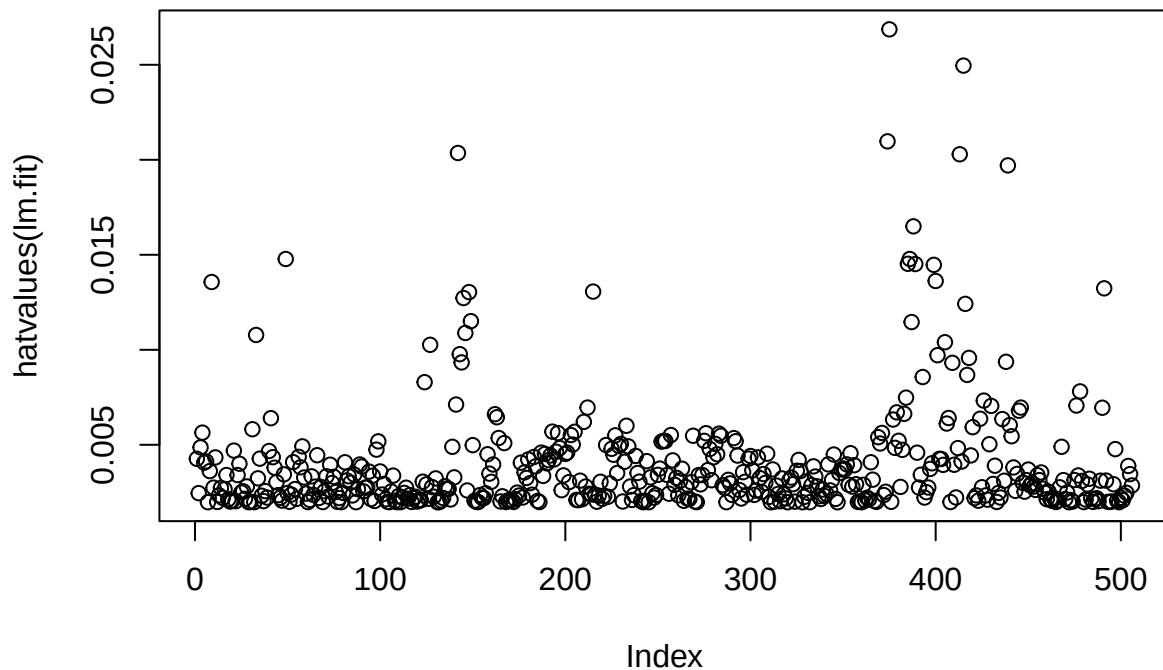


```
plot(predict(lm.fit), rstudent(lm.fit))
```



Plot for residual plots

```
plot(hatvalues(lm.fit))
```



```
which.max(hatvalues(lm.fit))
```

```
## 375
## 375
```

Multiple Linear Regression

```
lm.fit <- lm(medv ~ lstat + age, data = Boston)
summary(lm.fit)
```

```
##
## Call:
## lm(formula = medv ~ lstat + age, data = Boston)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
##	-15.981	-3.978	-1.283	1.968	23.158

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	33.22276	0.73085	45.458	< 2e-16 ***
## lstat	-1.03207	0.04819	-21.416	< 2e-16 ***

```
## age          0.03454    0.01223    2.826  0.00491 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.173 on 503 degrees of freedom
## Multiple R-squared:  0.5513, Adjusted R-squared:  0.5495
## F-statistic: 309 on 2 and 503 DF, p-value: < 2.2e-16
```

Multiple regression of all predictors

```
lm.fit <- lm(medv ~ ., data = Boston)
summary(lm.fit)
```

```
##
## Call:
## lm(formula = medv ~ ., data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1304  -2.7673  -0.5814   1.9414  26.2526
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  41.617270   4.936039   8.431 3.79e-16 ***
## crim        -0.121389   0.033000  -3.678 0.000261 ***
## zn           0.046963   0.013879   3.384 0.000772 ***
## indus        0.013468   0.062145   0.217 0.828520
## chas         2.839993   0.870007   3.264 0.001173 **
## nox        -18.758022   3.851355  -4.870 1.50e-06 ***
## rm           3.658119   0.420246   8.705 < 2e-16 ***
## age          0.003611   0.013329   0.271 0.786595
## dis        -1.490754   0.201623  -7.394 6.17e-13 ***
## rad          0.289405   0.066908   4.325 1.84e-05 ***
## tax         -0.012682   0.003801  -3.337 0.000912 ***
## ptratio     -0.937533   0.132206  -7.091 4.63e-12 ***
## lstat       -0.552019   0.050659 -10.897 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.798 on 493 degrees of freedom
## Multiple R-squared:  0.7343, Adjusted R-squared:  0.7278
## F-statistic: 113.5 on 12 and 493 DF, p-value: < 2.2e-16
```

VIF

```
vif(lm.fit)
```

```
##      crim      zn      indus      chas      nox      rm      age      dis
```

```
## 1.767486 2.298459 3.987181 1.071168 4.369093 1.912532 3.088232 3.954037
##      rad      tax ptratio  lstat
## 7.445301 9.002158 1.797060 2.870777
```

Running model with no un significant variable

```
lm.fit1 <-lm(medv~.- age, data = Boston)
summary(lm.fit1)
```

```
##
## Call:
## lm(formula = medv ~ . - age, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1851  -2.7330  -0.6116   1.8555  26.3838
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  41.525128   4.919684   8.441 3.52e-16 ***
## crim        -0.121426   0.032969  -3.683 0.000256 ***
## zn           0.046512   0.013766   3.379 0.000785 ***
## indus        0.013451   0.062086   0.217 0.828577
## chas         2.852773   0.867912   3.287 0.001085 **
## nox        -18.485070   3.713714  -4.978 8.91e-07 ***
## rm           3.681070   0.411230   8.951 < 2e-16 ***
## dis        -1.506777   0.192570  -7.825 3.12e-14 ***
## rad          0.287940   0.066627   4.322 1.87e-05 ***
## tax         -0.012653   0.003796  -3.333 0.000923 ***
## ptratio     -0.934649   0.131653  -7.099 4.39e-12 ***
## lstat       -0.547409   0.047669 -11.483 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.794 on 494 degrees of freedom
## Multiple R-squared:  0.7343, Adjusted R-squared:  0.7284
## F-statistic: 124.1 on 11 and 494 DF,  p-value: < 2.2e-16
```

or use update ()

```
lm.fit1 <-update(lm.fit,~.- age)
summary(lm.fit1)
```

```
##
## Call:
## lm(formula = medv ~ crim + zn + indus + chas + nox + rm + dis +
##      rad + tax + ptratio + lstat, data = Boston)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1851  -2.7330  -0.6116   1.8555  26.3838
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  41.525128   4.919684   8.441 3.52e-16 ***
## crim        -0.121426   0.032969  -3.683 0.000256 ***
## zn           0.046512   0.013766   3.379 0.000785 ***
## indus        0.013451   0.062086   0.217 0.828577
## chas         2.852773   0.867912   3.287 0.001085 **
## nox        -18.485070   3.713714  -4.978 8.91e-07 ***
## rm           3.681070   0.411230   8.951 < 2e-16 ***
## dis        -1.506777   0.192570  -7.825 3.12e-14 ***
## rad          0.287940   0.066627   4.322 1.87e-05 ***
## tax         -0.012653   0.003796  -3.333 0.000923 ***
## ptratio     -0.934649   0.131653  -7.099 4.39e-12 ***
## lstat       -0.547409   0.047669 -11.483 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.794 on 494 degrees of freedom
## Multiple R-squared:  0.7343, Adjusted R-squared:  0.7284
## F-statistic: 124.1 on 11 and 494 DF,  p-value: < 2.2e-16
```

InteractionTerms

```
summary(lm(medv~lstat * age, data = Boston))
```

```
##
## Call:
## lm(formula = medv ~ lstat * age, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.806  -4.045  -1.333   2.085  27.552
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  36.0885359   1.4698355  24.553 < 2e-16 ***
## lstat       -1.3921168   0.1674555  -8.313 8.78e-16 ***
## age         -0.0007209   0.0198792  -0.036  0.9711
## lstat:age     0.0041560   0.0018518   2.244  0.0252 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.149 on 502 degrees of freedom
## Multiple R-squared:  0.5557, Adjusted R-squared:  0.5531
## F-statistic: 209.3 on 3 and 502 DF,  p-value: < 2.2e-16
```


Non-linear Transformations of the Predictors

```
lm.fit2 <-lm(medv~lstat + I(lstat^2))
summary(lm.fit2)
```

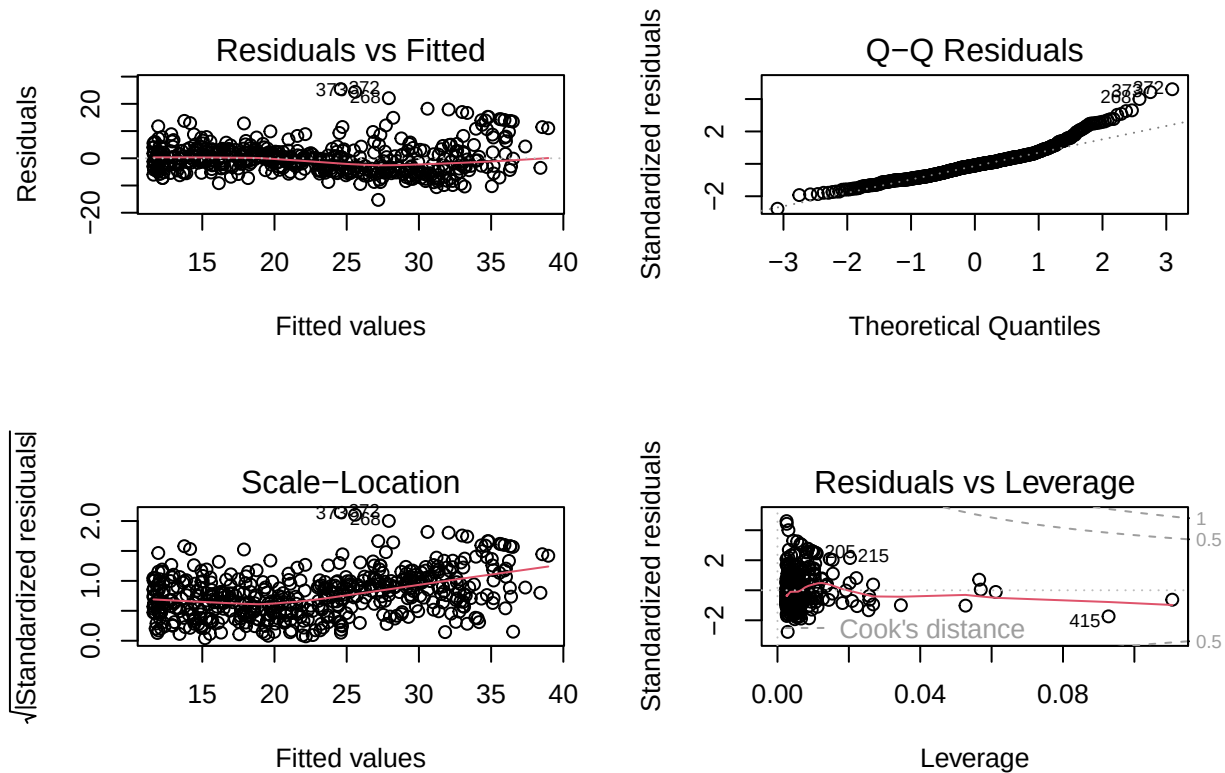
```
##
## Call:
## lm(formula = medv ~ lstat + I(lstat^2))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.2834  -3.8313  -0.5295   2.3095  25.4148
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 42.862007   0.872084  49.15   <2e-16 ***
## lstat       -2.332821   0.123803 -18.84   <2e-16 ***
## I(lstat^2)   0.043547   0.003745  11.63   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.524 on 503 degrees of freedom
## Multiple R-squared:  0.6407, Adjusted R-squared:  0.6393
## F-statistic: 448.5 on 2 and 503 DF,  p-value: < 2.2e-16
```

The near-zero p-value associated with the quadratic term suggests that it leads to an improved model

```
lm.fit <-lm(medv~lstat)
anova(lm.fit, lm.fit2)
```

```
## Analysis of Variance Table
##
## Model 1: medv ~ lstat
## Model 2: medv ~ lstat + I(lstat^2)
##   Res.Df  RSS Df Sum of Sq    F    Pr(>F)
## 1     504 19472
## 2     503 15347   1    4125.1 135.2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
par(mfrow = c(2, 2))
plot(lm.fit2)
```



using the `poly()` function `poly()`

#to create the polynomial within `lm()`

```
lm.fit5 <- lm(medv ~ poly(lstat, 5))
summary(lm.fit5)
```

```
##
## Call:
## lm(formula = medv ~ poly(lstat, 5))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.5433  -3.1039  -0.7052   2.0844  27.1153
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    22.5328    0.2318  97.197 < 2e-16 ***
## poly(lstat, 5)1 -152.4595    5.2148 -29.236 < 2e-16 ***
```

```
## poly(lstat, 5)2    64.2272    5.2148  12.316 < 2e-16 ***
## poly(lstat, 5)3   -27.0511    5.2148  -5.187 3.10e-07 ***
## poly(lstat, 5)4    25.4517    5.2148   4.881 1.42e-06 ***
## poly(lstat, 5)5   -19.2524    5.2148  -3.692 0.000247 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.215 on 500 degrees of freedom
## Multiple R-squared:  0.6817, Adjusted R-squared:  0.6785
## F-statistic: 214.2 on 5 and 500 DF,  p-value: < 2.2e-16
```

```
summary(lm(medv ~ log(rm), data = Boston))
```

```
##
## Call:
## lm(formula = medv ~ log(rm), data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.487  -2.875  -0.104   2.837  39.816
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -76.488     5.028  -15.21 <2e-16 ***
## log(rm)         54.055     2.739   19.73 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.915 on 504 degrees of freedom
## Multiple R-squared:  0.4358, Adjusted R-squared:  0.4347
## F-statistic: 389.3 on 1 and 504 DF,  p-value: < 2.2e-16
```

Qualitative predictors

```
head(Carseats)
```

```
##   Sales CompPrice Income Advertising Population Price ShelveLoc Age Education
## 1  9.50      138     73           11         276   120        Bad   42         17
## 2 11.22      111     48           16         260    83        Good   65         10
## 3 10.06      113     35           10         269    80       Medium   59         12
## 4  7.40      117    100            4         466    97       Medium   55         14
## 5  4.15      141     64            3         340   128        Bad   38         13
## 6 10.81      124    113           13         501    72        Bad   78         16
##   Urban  US
## 1   Yes  Yes
## 2   Yes  Yes
```

```
## 3   Yes Yes
## 4   Yes Yes
## 5   Yes  No
## 6    No Yes
```

```
lm.fit <- lm(Sales ~ . + Income:Advertising + Price:Age,
data = Carseats)
summary(lm.fit)
```

```
##
## Call:
## lm(formula = Sales ~ . + Income:Advertising + Price:Age, data = Carseats)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.9208 -0.7503  0.0177  0.6754  3.3413
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.5755654   1.0087470    6.519 2.22e-10 ***
## CompPrice      0.0929371   0.0041183   22.567 < 2e-16 ***
## Income         0.0108940   0.0026044    4.183 3.57e-05 ***
## Advertising    0.0702462   0.0226091    3.107 0.002030 **
## Population     0.0001592   0.0003679    0.433 0.665330
## Price         -0.1008064   0.0074399  -13.549 < 2e-16 ***
## ShelveLocGood  4.8486762   0.1528378   31.724 < 2e-16 ***
## ShelveLocMedium 1.9532620   0.1257682   15.531 < 2e-16 ***
## Age           -0.0579466   0.0159506   -3.633 0.000318 ***
## Education     -0.0208525   0.0196131   -1.063 0.288361
## UrbanYes       0.1401597   0.1124019    1.247 0.213171
## USYes         -0.1575571   0.1489234   -1.058 0.290729
## Income:Advertising 0.0007510  0.0002784    2.698 0.007290 **
## Price:Age      0.0001068  0.0001333    0.801 0.423812
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.011 on 386 degrees of freedom
## Multiple R-squared:  0.8761, Adjusted R-squared:  0.8719
## F-statistic: 210 on 13 and 386 DF, p-value: < 2.2e-16
```

```
attach(Carseats)
contrasts(ShelveLoc)
```

```
##           Good Medium
```

```
## Bad      0      0
## Good     1      0
## Medium   0      1
```

writing a function

```
LoadLibraries <-function() {
  library(ISLR2)
  library(MASS)
  print("The libraries have been loaded.")
}
```

```
LoadLibraries()
```

```
## [1] "The libraries have been loaded."
```